

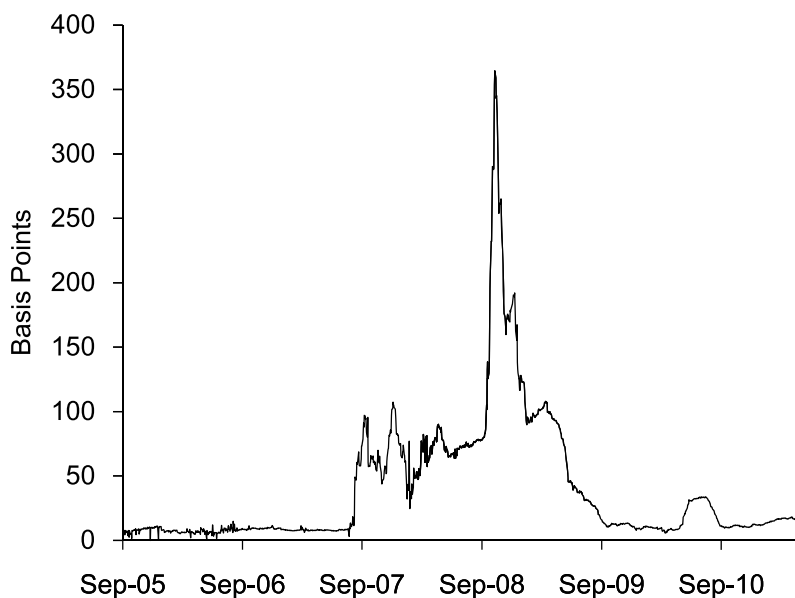
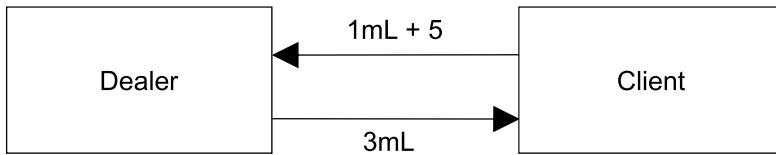


Figure 1.7: Historical LIBOR-OIS Spread

1.2.2 Basis Swaps

In a basis swap, both streams of payment are floating in nature. Whereas the fixed coupon in a fixed-floating swap is set such that the trade has zero value today, in a basis swap a spread is added to one of the floating legs in order to make it an $NPV = 0$ transaction.

As an example, consider the swap in Figure 1.8 that shows a *1s3s* (pronounced “ones threes”) basis swap, a swap in which one counterparty pays 1-month LIBOR, and the other pays 3-month LIBOR. Notice that the dealer is agreeing to receive 1-month LIBOR plus 5 bps monthly money and pay 3-month LIBOR quarterly money for two years. In a *1s3s* basis swap, when the dealer receives 1-month LIBOR plus a spread, it is referred to as the offer side of the basis swap. The bid side of the basis swap would have the dealer paying 1-month LIBOR plus a (lower) spread monthly money. Since the dealer is the one making a market in the basis swap, the bid side will have a spread that is lower than the offer side. Thus, for example, the dealer may be willing to pay 1-month LIBOR plus 3 bps versus receiving 3-month LIBOR for two years (on the bid side) and may be willing to receive 1-month LIBOR plus 5 bps versus paying 3-month LIBOR for two years (on the offer side). Simply put, the *1s3s* basis swap is “bid at 3, offered at 5.” The mid-



Client pays: 1-month LIBOR + 5 bps ("1mL + 5") monthly on Act/360 basis
 Dealer pays: 3-month LIBOR ("3mL") quarterly on Act/360 basis
 Notional: \$100mm
 Maturity: 2 years

Figure 1.8: A "1s3s" Basis Swap

market in the basis swap would be defined as the average of the bid and the offer, or a spread of 4 bps in this case.

Why is there a spread at all in this basis swap? That is, why isn't the dealer willing to receive 1-month LIBOR *flat*, or plus 0 bps, versus 3-month LIBOR? The single biggest determinant of the spread in the 1s3s basis swap is supply and demand: as investors want to pay 1-month LIBOR in the basis swap, that will increase the spread, and vice versa. But aside from this, there are other differences in the two legs of the basis swap that would suggest the spread should be nonzero. First, assuming the basis swap is spot starting, the initial 1-month and 3-month LIBOR settings are known with certainty and are typically not the same. For example, in an upward-sloping yield curve environment, it is usually the case that 3-month LIBOR is greater than 1-month LIBOR, suggesting a factor that would make the spread positive. These are of course only the first settings in the swap (of the 24 settings on the 1-month LIBOR side and the 8 settings on the 3-month LIBOR side). If the market's expectation is that 1-month LIBOR will be lower than 3-month LIBOR over the next two years, that would also suggest that the spread in the basis swap should be positive. Finally, payments on the 1-month LIBOR side occur monthly, whereas payments on the 3-month LIBOR side occur quarterly. Since it is always better to receive money sooner rather than later, this factor in isolation would actually suggest that the spread should be negative. Taking into account all of these factors, the mid-market spread for a 1s3s basis swap for two years is 4 bps.

Figure 1.9 shows a history of the 1s3s basis swap for a maturity of two years. As the figure demonstrates, the basis swap spread was not particularly volatile prior to the financial crisis. But during the financial crisis, the basis swap spread experienced tremendous volatility and widened out significantly. As the crisis was mounting, so were concerns about liquidity. There

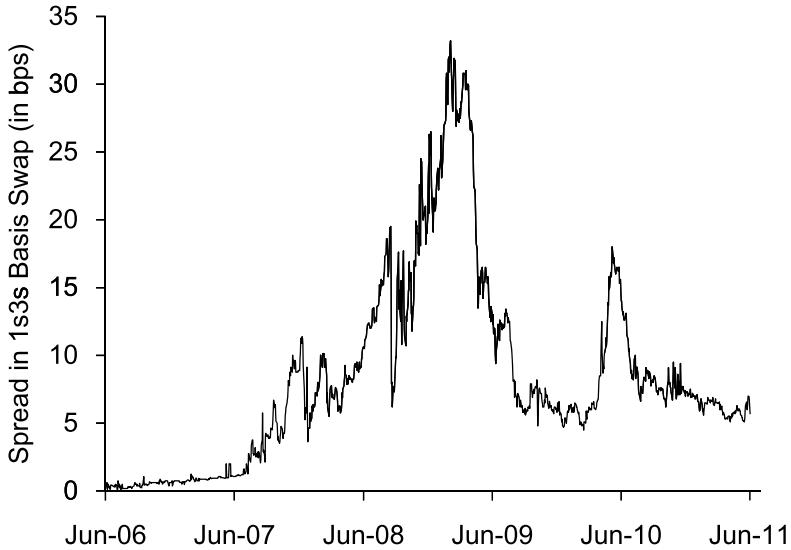


Figure 1.9: History of “1s3s” Basis Swap, 2-Year Maturity

were concerns abounding that financial institutions were reluctant to lend, particularly for longer periods of time. Many in the market expected that financial institutions would be less willing to lend for, say, three months than to lend for, say, one month. This would manifest itself in a wider differential between 3-month LIBOR and 1-month LIBOR. The expectation in the market was that this wider differential would persist for some time, and this was reflected by a widening of the basis swap spread.

Example

Let’s look back at Table 1.1 and suppose that a swap dealer is willing to make a *two-way market* in 2-year swaps as indicated on the screen.⁴⁰ That is, the dealer is willing to pay 0.665% semi bond versus threes and is willing to receive 0.705% semi bond versus threes. Suppose that at the same time

⁴⁰In the swaps market, it was common etiquette for many years that if a client called a dealer and wanted to do a swap, the client would indicate first if he were interested in paying fixed or receiving fixed prior to asking for a swap rate from a dealer. Thus, a client might say something like, “Where can I pay fixed on \$100mm 5-year swaps?” For many years, asking for a two-way market (i.e., asking a dealer to simultaneously show a bid side and offer side swap rate and expecting the dealer to be prepared to deal at either level) was considered a huge breach of etiquette. In more recent years, it has become acceptable to ask a dealer to show a two-way.

the dealer is a 3 bid and a 5 offer in a 1s3s basis swap for two years. That is, the dealer is willing to pay 1-month LIBOR + 3 bps monthly money versus 3-month LIBOR quarterly money and is also willing to receive 1-month LIBOR + 5 bps monthly money versus 3-month LIBOR quarterly money, both for a term of two years. If a client is interested in receiving fixed semi bond versus 1-month LIBOR paid monthly money for two years, what fixed rate should he expect to see from the dealer?

Let's look at this trade from the point of view of the swap dealer. The dealer is being asked to pay fixed for two years. We know that the dealer is willing to pay fixed at 0.665% semi bond. However, this is the rate the dealer is willing to pay versus 3-month LIBOR quarterly money. The dealer is, however, being asked to receive 1-month LIBOR monthly money instead. As we have discussed, any change to the floating leg from the standard 3-month LIBOR implies that the fixed rate in the swap will change. The swap dealer is not indifferent between receiving 3-month LIBOR quarterly money for two years and receiving 1-month LIBOR *flat* (meaning plus a spread of 0 bps) monthly money for two years. We know this from the basis swap: in particular, we know that if the dealer is receiving 1-month LIBOR, he will want to receive 1-month LIBOR + 5 bps monthly money versus paying 3-month LIBOR quarterly money for two years.

As shown in Figure 1.10, the implication of this is that the dealer will be willing to pay 0.615% semi bond versus 1-month LIBOR monthly money. We can think about decomposing a swap in which a dealer pays fixed semi bond versus ones into two swaps: a standard swap in which a dealer pays fixed versus threes and a 1s3s basis swap in which a dealer receives 1-month LIBOR and pays 3-month LIBOR. We know that the dealer is willing to pay 0.665% versus threes, and we also know that a dealer is willing to receive 1-month LIBOR + 5 monthly money versus 3-month LIBOR quarterly money. Each of these two transactions is shown in the decomposition of the swap into two pieces. Notice that the two 3-month LIBOR legs in the decomposition are offsetting and thus cancel each other out. We are left with the dealer paying fixed 0.665% versus receiving 1-month LIBOR + 5 monthly money. The only problem is that the client wants to pay 1-month LIBOR flat, not 1-month LIBOR + 5, in the terms of the trade. However, if we “subtract” 5 bps from each leg of the swap, we see that the dealer is willing to receive $0.665\% - 0.05\% = 0.615\%$ versus 1-month LIBOR.⁴¹ $\square$

⁴¹We are free to “subtract” 5 bps from each leg of the swap because doing so does not change the value of the swap. We should be careful, however, because the 5 bps in the 1s3s basis swap is quoted on a monthly money basis, whereas the fixed leg of the swap is semi bond. Thus, subtracting the 5 bps from the fixed rate is technically wrong. We should convert the 5 bps paid monthly money to an appropriate number of bps paid semi bond. The magnitude of this error is so small as to be safely ignored for our purposes here.

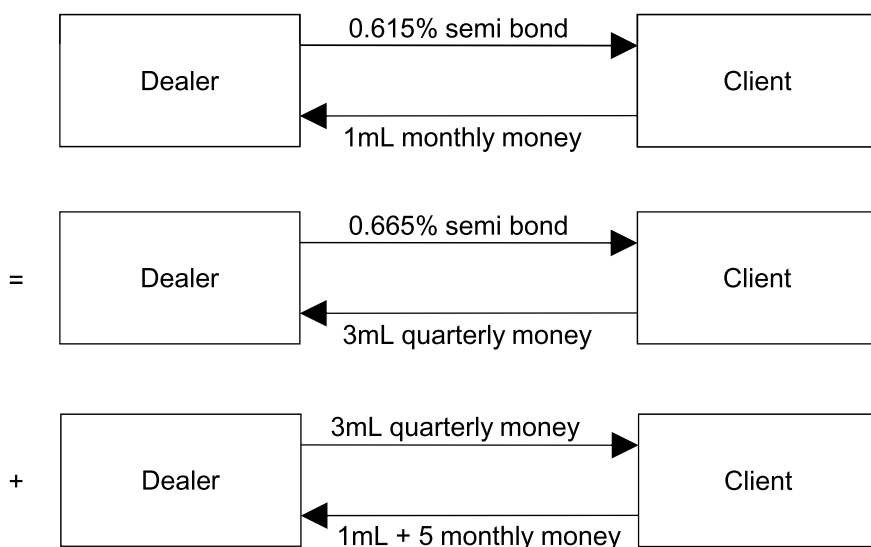


Figure 1.10: Decomposing a Fixed-Floating Swap Versus 1-Month LIBOR

Let's consider now another type of basis swap as shown in Figure 1.11. The first thing to note about this transaction is that it is a *forward starting swap*, meaning it is a swap transacted at one point in time with an effective date beyond the normal $T + 2$ settlement. In this case, the swap is a "2x5" (pronounced "2 by 5"), meaning a 3-year swap, two years forward. The swap itself will last a total of three years. The effective date of the swap is two years hence. First payments on each side are in 2.25 years, and final payments are five years from today, which is three years after the effective date of the swap.

In this basis swap, the dealer is receiving 2-year CMS, or *Constant Maturity Swap*. The 2-year CMS rate is a floating index that is set equal to the 2-year swap rate each day at exactly 11:00 A.M. in New York. CMS rates for other maturities are similarly defined. The convention in the market is that derivative transactions that depend on various CMS rates refer to a data source known as ISDAFIX (which was created by ISDA). ISDAFIX displays swap rates of various maturities for a variety of currencies each day. These rates are also displayed in the Federal Reserve's Statistical Release H.15.⁴²

⁴²For USD swaps, ISDAFIX actually makes rates available from a mid-day polling at 11:00 A.M. New York time as well as an end-of-day polling at 3:00 P.M. New York time. A number of dealers are polled and asked to submit the mean of where the dealer would bid

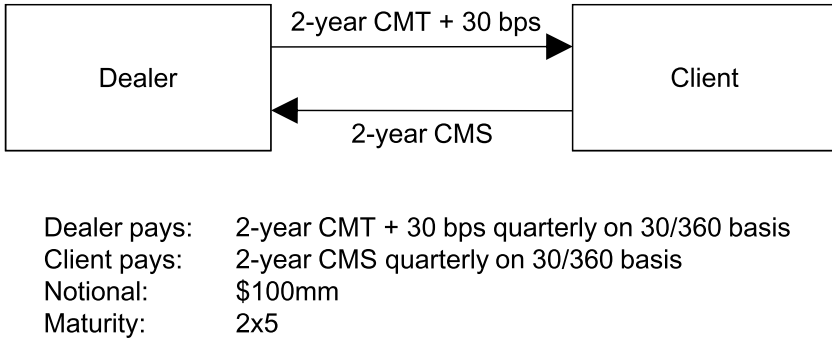


Figure 1.11: A CMS-CMT Basis Swap

The dealer is paying a floating rate of interest equal to 2-year CMT, or *Constant Maturity Treasury*, plus 30 bps. Constant Maturity Treasury rates are yields on nominal Treasury issues adjusted to constant maturities, as posted in the Federal Reserve Statistical Release H.15. We can think of 2-year CMT as the daily rate on a hypothetical Treasury issue with 2 years until maturity.⁴³ On each of the 12 quarterly resets (the first of which is in two years), levels for 2-year CMT and 2-year CMS are set. The spread of 30 is added to the 2-year CMT setting, which is paid by the dealer at the end of each quarter in exchange for the 2-year CMS setting.

How do we interpret this trade? You may recall from Section 1.2.1 that a swap spread is the difference between the swap rate of a given maturity and the yield of the respective on-the-run Treasury. Since this basis swap has no upfront cash flows, one (loose) interpretation of this trade is that the 2-year swap spread over the three-year period commencing two years hence is, on average, expected to be 30 bps.⁴⁴ Which side of this trade would we want

and offer a swap of a particular maturity for a notional of \$50mm. Of the quotes obtained, the four highest and four smallest are tossed, and a simple average of the remaining quotes constitutes the rate setting. See <http://www.isda.org/fix/isdafix.html> for more details.

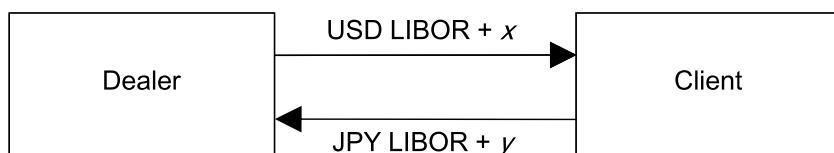
⁴³The 2-year CMT rate does not, in general, exactly equal the yield of the on-the-run 2-year Treasury. Whereas the 2-year swap rate observed each day has a maturity of exactly two years, the 2-year on-the-run Treasury does not have two years maturity in general. A methodology is employed to adjust the yields on nominal Treasuries to a constant maturity of two years, and this yield is the 2-year CMT rate.

⁴⁴Due to some technicalities that we need not concern ourselves with here, this loose interpretation is not entirely accurate. The risk profile of this trade is more complicated than this. In addition, as we will see in Chapter 6, trades like these have some option-like features to them.

1. Initial Principal Exchange:

Dealer pays JPY 8.53bn; receives USD 100mm

2. Interest Flows:



3. Final Principal Exchange:

Dealer pays USD 100mm; receives JPY 8.53bn

Figure 1.12: A Cross-Currency Swap

to take if we thought that 2-year swap spreads in the “2 by 5” period were going to be, on average, less than 30 bps?⁴⁵

1.2.3 Cross-Currency Swaps

A *cross-currency swap* is a swap in which the streams of payment over the life of the trade are made in two different currencies. The streams of payment can both be fixed, both be floating (in which case the swap is a *cross-currency basis swap*) or consist of one fixed stream and one floating stream. The two counterparties in a cross-currency swap typically exchange specific amounts of the two different currencies at the beginning and end of the trade. Those amounts are determined by the spot exchange rate between the two currencies.

Example

Suppose that the spot \$/yen exchange rate is 85.3, which is to say that U.S. \$1 can be exchanged for 85.3 Japanese yen (JPY). Consider the transaction shown in Figure 1.12.

⁴⁵We would want to take the position of the client. Let’s say that 2-year swap spreads each quarter during the “2x5” period were 20 bps. Then the client would wind up receiving 2-year CMT + 30 and paying a stream of payments that is (roughly) 10 bps lower.

Notice that at the spot exchange rate of 85.3, \$100mm can be exchanged for 8.53bn JPY. These are the amounts that are exchanged not only on the effective date of the swap but also on the maturity date. Regardless of where \$/yen exchange rates may be at maturity, the final exchange of principal is based solely on where the exchange rate is when the trade is executed.

One way to interpret this trade is as follows: the dealer is receiving \$100mm on the effective date and is financing this “dollar loan” over the life of the trade by paying USD LIBOR + x bps. Simultaneously, the client is receiving 8.53bn yen on the effective date and is financing this “yen loan” by paying JPY LIBOR + y bps. The quantities x and y are set such that the trade has zero net present value at the inception of the trade. $\square$

Problem Set 1

Questions 1 and 2 below make reference to the following composite page for swaps:

Maturity	Treasury Prices	Treasury Yields	Mid Yields	Swap Spreads
2 Year	100-042/100-04+	0.557/0.553	0.555	19/18.5
3 Year	100-016/100-02+	0.981/0.979	0.980	24.5/24
5 Year	100-18+/100-186	1.877/1.876	1.877	21.25/20.75
7 Year	100-14/100-146	2.556/2.552	2.554	17/16.5
10 Year	99-04/99-04+	3.228/3.226	3.227	7.5/7
30 Year	106-20+/106-21	4.350/4.349	4.350	-24.5/-25

1. Swap rates for all other maturities that are not explicitly shown on this screen are computed as a spread to the linearly interpolated Treasury yield, making use of the mid-market yields of the two adjacent on-the-run Treasuries.

a) If the market for 4-year swap spreads is 25.25/24.75, what is the bid side swap rate and offer side swap rate for four years?

b) If the market for 15-year swap spreads is 27.5/27, what is the bid side swap rate and offer side swap rate for 15 years?

2. Referring to the above composite page:

a) What is the 5-year offer side swap rate quoted semi bond? Be sure to use the appropriate side of the Treasury and swap spread when answering this question.

b) Based on your answer to part a), what is the 5-year offer side swap rate quoted quarterly money?

c) Suppose a client would like to receive fixed in a 10-year swap. At what rate should he expect to get done if he expects to get done “at the screen”? That is, suppose that the client expects that he can get done at a rate that reflects the appropriate bid/offer levels as indicated on the screen. At what level does he expect to get done?

d) This question is a continuation of part c). Suppose that the dealer indicates that it can transact at a rate that is more advantageous to the client. If the dealer can improve the level to the client by 0.5 bps, what level should the client expect to see?

3. Suppose that the following information about basis swaps is made available by a swap dealer:

	1s3s LIBOR Basis Swaps		Prime-LIBOR Basis Swaps	
	Bid	Offer	Bid	Offer
1 Year	+3	+5	−283	−279
2 Year	+2	+4	−281	−277
3 Year	+1	+3	−280	−276
4 Year	+1	+3	−278	−274
5 Year	+1	+3	−276	−272

These basis swap spreads are quoted as a spread to the non-3-month LIBOR side of the basis swap. Thus, for example, the swap dealer is willing to pay Prime minus 276 bps in order to receive 3-month LIBOR for five years, and the dealer is also willing to receive Prime minus 272 bps in order to pay 3-month LIBOR for five years.

In a 1s3s basis swap, the standard convention is that 1-month LIBOR resets once at the beginning of each month, accrues, and is paid at the end of the month on a monthly Act/360 basis. In a Prime-LIBOR basis swap, Prime, which is the interest rate that banks charge to its middle-market customers, is reset daily during each day during a quarter. At the end of a quarter, these daily Prime rates observed during each day of the quarter are averaged, and that average is used for the quarterly rate set on the Prime side.⁴⁶ That average is then paid on a quarterly Act/360 basis. In all cases, 3-month LIBOR is reset once at the beginning of each quarter, accrues, and is paid at the end of the quarter on an Act/360 basis, as is standard.

Suppose further that the following information about swaps rates is made available from a swap dealer (these rates are quoted semi 30/360 versus 3-month LIBOR):

⁴⁶If you want to get really technical, Prime is computed on a daily weighted average, meaning that a rate setting on Friday counts for Saturday and Sunday as well.